

Technical Requirement Document

Here is the Technical Requirement Document (TRD) based on the provided Functional Requirement Document (FRD):

TR-DLT-001: Load Customer and Order Data into Delta Tables

Related Functional Requirement(s): FR-DLT-001

Objective: Implement a Delta Live Table pipeline to load customer and order data from CSV files into Delta tables.

Target Cluster Configuration:

- Cluster Name: Gen AI POC Databricks Cluster
- Databricks Runtime Version: 12.2 LTS (or latest)
- Node Type: Standard_DS3_v2
- Driver Node: Standard_DS3_v2
- Worker Nodes: 2-5 nodes, depending on autoscaling
- Autoscaling: Enabled
- Auto Termination: 30 minutes
- Libraries Installed: delta-lake, pyspark

Source Data Details:

- Customer CSV data: `/Volumes/gen\_ai\_poc\_databrickscOE/sdlc\_wizard/customerdata`, CSV, Customer data
- Order CSV data: `/Volumes/gen\_ai\_poc\_databrickscOE/sdlc\_wizard/orderdata`, CSV, Order data

Target Data Details:

- Customer Delta table: `customer`, Delta, Customer data
- Order Delta table: `order`, Delta, Order data

Job Flow / Pipeline Stages:

- Read customer CSV data into a Delta table
- Read order CSV data into a Delta table
- Add a new column "TotalAmount" to the "order" Delta table
- Remove null and duplicate records from both Delta tables

Data Transformations / Business Logic:

- Step 1: Read CSV data into DataFrames
- Step 2: Write DataFrames to Delta tables
- Step 3: Add "TotalAmount" column to "order" Delta table using `PricePerUnit` and `Qty` columns

- Step 4: Remove null and duplicate records using ``dropDuplicates()`` and ``dropna()``

Error Handling and Logging:

- Log errors to a Delta table or a logging framework
- Handle null and duplicate records by removing them

TR-DLT-002: Create and Load Order Summary Table

Related Functional Requirement(s): FR-DLT-002

Objective: Implement a Delta Live Table pipeline to create and load the "ordersummary" table with joined customer and order data using SCD type 2.

Target Cluster Configuration: Same as TR-DLT-001

Source Data Details:

- Customer Delta table: ``customer``, Delta, Customer data
- Order Delta table: ``order``, Delta, Order data

Target Data Details:

- Order Summary Delta table: ``ordersummary``, Delta, Joined customer and order data

Job Flow / Pipeline Stages:

- Create the "ordersummary" table if it does not exist
- Join the "customer" and "order" Delta tables on the "CustId" field
- Load the joined data into the "ordersummary" table using SCD type 2

Data Transformations / Business Logic:

- Step 1: Join "customer" and "order" Delta tables using ``join()``
- Step 2: Apply SCD type 2 logic using ``merge()`` and ``update()``

Error Handling and Logging: Same as TR-DLT-001

TR-DLT-003: Update Order Summary Table on Customer Changes

Related Functional Requirement(s): FR-DLT-003

Objective: Implement a Delta Live Table pipeline to update the "ordersummary" table whenever there is a change in the "customer" table.

Target Cluster Configuration: Same as TR-DLT-001

Source Data Details:

- Customer Delta table: ``customer``, Delta, Customer data

Target Data Details:

- Order Summary Delta table: ``ordersummary``, Delta, Joined customer and order data

Job Flow / Pipeline Stages:

- Identify changes made to the "customer" table
- Update the "ordersummary" table by making old records inactive and new records active
- Update the "StartDate" and "EndDate" fields accordingly to maintain history

Data Transformations / Business Logic:

- Step 1: Identify changes using ``exceptAll()`` and ``join()``
- Step 2: Update "ordersummary" table using ``merge()`` and ``update()``

Error Handling and Logging: Same as TR-DLT-001

TR-DLT-004: Create and Load Customer Aggregate Spend Table

Related Functional Requirement(s): FR-DLT-004

Objective: Implement a Delta Live Table pipeline to create and load the "customeraggregatespend" table with aggregated data from the "ordersummary" table.

Target Cluster Configuration: Same as TR-DLT-001**Source Data Details:**

- Order Summary Delta table: ``ordersummary``, Delta, Joined customer and order data

Target Data Details:

- Customer Aggregate Spend Delta table: ``customeraggregatespend``, Delta, Aggregated customer spend data

Job Flow / Pipeline Stages:

- Create the "customeraggregatespend" table if it does not exist
- Aggregate the "TotalAmount" column from the "ordersummary" table, grouping by "Name" and "Date"
- Load the aggregated data into the "customeraggregatespend" table

Data Transformations / Business Logic:

- Step 1: Aggregate data using ``groupBy()`` and ``sum()``

Error Handling and Logging: Same as TR-DLT-001

TR-DLT-005: Implement Delta Live Tables

Related Functional Requirement(s): FR-DLT-005

Objective: Implement the above requirements using Delta Live Tables.

Target Cluster Configuration: Same as TR-DLT-001**Job Flow / Pipeline Stages:**

- Implement the loading of customer and order data into Delta tables using Delta Live Tables

- Implement the creation and loading of the "ordersummary" table using Delta Live Tables
- Implement the update of the "ordersummary" table on customer changes using Delta Live Tables
- Implement the creation and loading of the "customeraggregatespend" table using Delta Live Tables

Data Transformations / Business Logic: Same as above TRs

Error Handling and Logging: Same as TR-DLT-001